

FACILITY REQUIREMENT

No.	PROCESSING STEP	REQUIREMENT	DETAILS
1	Machine Size	Length	1574mm
		Width	1674mm
		Height	1346mm
		Weight	1800kg
2	Machine Incoming Supply	Volt :	3phase 220V
		Amp:	50 Amp
		Hertz	50 ~ 60hz
3	Dust Collector Unit	Dimension	Decide by Maruwa
		Voltage/Amp	Decide by Maruwa
		Flow rate	150 CFM
		Ducting size	3 1/4" (OD)
4	Chiller Unit	Dimension	'MARUWA UNIT'
		Voltage/Amp	'MARUWA UNIT'
		Flow rate	11ltr/min
		Piping Size	1/2" Male Pipe
5	Operation Gas Facility	Gas piping size	No required due to enclosed system
		Types	
6	Compress air	Clean Air ? Normal Air?	Clean air
		Filtration?	Required
		Pressure?	30psi (depends on laser diameter)
7	Wear & Tear Parts	Part List & Drawing All parts	Lens, Filter & Table greasing (min 6 months)
8	Humidity Control	Must maintain dry condition in laser head	Nitrogen with 3psi

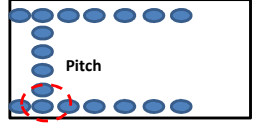
PROCESS CONTROL

No.	PROCESSING STEP	UNIT	PROCESS CONTROL	REQUIREMENT	RECORDS	ACTION	EDI EXPLANATION
1	Fresh CO2 gas is supplied	Gases	CW Gas Pressure 18 ~ 21 mbar	Digital control & Numerical value display	Data capturing (LOG)	Machine automatic stop & need to verify part replace person	Not Applicable. Totally different system and this is the close type system where the gases is consiled in
			N2: 75 ~ 85kgf/cm				
			He: 430 ~ 480kgf/cm				
			REP Gas Pressure 7 ~ 11 mbar				
			CO2 : 40 ~ 50kgf/cm				
			N2: 14 ~ 18kgf/cm				
			He: 180 ~ 200kgf/cm				
2	Proper voltage and current are supplied	Discharge electricity	Volt : 15 ~ 23 kV	Auto rotating brush cleaning & Proper Vacuum sunction	Cleaning & brush replacement counter		The brush will be installed and rotation can be done to ensure the nozzle being cleaned
			Amp: 10 ~ 40 mA				
3	Pumping of CO2 gas is done enough	Total reflector, output mirror	Dirty, stain	Digital control & Numerical value display	Data capturing (LOG)		The laser beam is consistent and laser power can be confirmed digitally by using the calibration gauge
			Nozzle cleaning, Lens Cleaning & change frequency				
4	The laser light power is constant and stable	Power meter	Power : 7 ~ 10 CW	Laser cutting measurement online (diameter)	Reflector cleaning counter		The laser alignment being judge by using ceramic piece and scribing it and just on the debris
5	The laser light reaches to the product without loss	Reflector	Laser Alignment - Arcylic Jig				
			Nozzle cleaning, Lens Cleaning & change frequency	Periodic table alignment check	Arcylic cutting confirmation????	Machine automatic stop & need to upload verification data	The machine being installed with linear table where the the LM guide changed to magnetic control Consistency for 100 cycle is 100%
6	Laser unit or table moves without warp	Movement unit	X & Y Table variances ($\pm 0.005\text{mm}$)				
			Test cut - circle diameter	Digital control & Numerical value display	Data capturing (LOG)	Machine automatic stop & need to verify part replace person	Cooling water rate should be in 113 ltr/min
7	Cooling water	Laser Unit	Power : 7 ~ 10 CW				



EDI LASER MACHINE - STANDARD REQUIRED SPEC & FEATURES

DEVICE CONTROL

NO	FAILURE CONTROL	IMPACT	REQUIREMENT	RECORDS	ACTION
1	Depth	Bari	ONLINE DEPTH & PITCH MEASUREMENT	Data capturing (LOG)	Mr Curt explain that the function of camera is for alignment only. Need to confirm on the camera's requirement  Pitch The scribing will be programmed where both axis will be met and have a proper breaking. Scribe depth is deeper at this point.
2	V Groove	Bari			
3	Scribe Line	Scribe Line			
4	Dimension	LL & WW Parallel, Angle	L & W MEASUREMENT BEFORE BREAK		

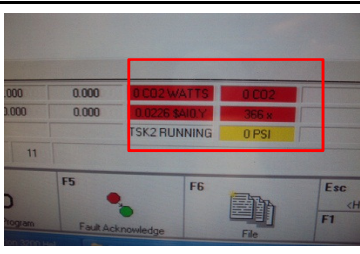
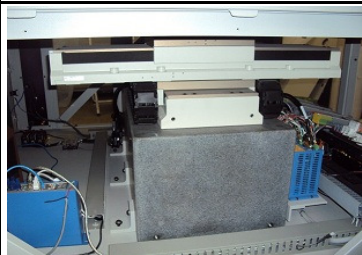
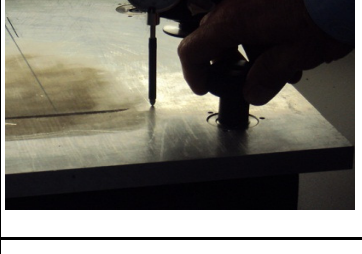
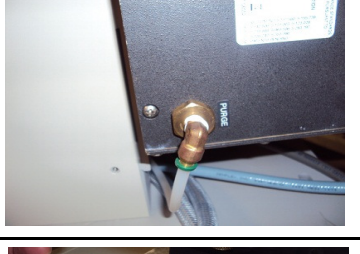
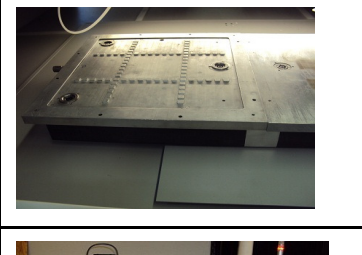
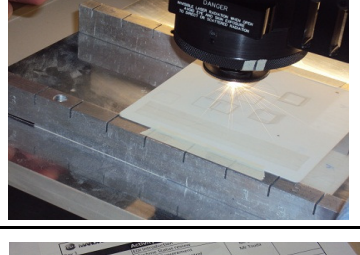
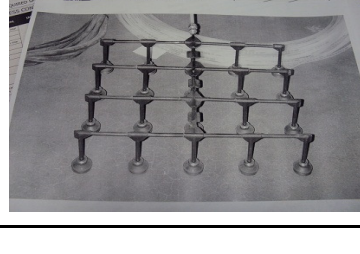
OVERALL CONTROL

NO	FAILURE CONTROL	IMPACT	REQUIREMENT	RECORDS	EDI STATUS/DETAILS
1	Admin access for overall control	Wrong operation	Pasword & Barcode detection		Mr Curt will discuss with his software company before answer
2	Table flatness	BARI	Calibration method		Will provide the equipment for calibrating the table
3	Work area material	Dirty	Teflon / Derlin Material	VCID System	Material used is Rubber mat (3M) and Nylon pin for stopper
4	Table lock/Product placement	Parallel	Product positioning		Plan to use vacum table instead of mechanical table
5	Ball screw override detetction	Dimension	Abnormal detection		Programme control linear table
6	CCD Camera position & Function		Camera is applied only for re-process or laser cut on printed substrate. For Maruwa Malaysia application, the camera have no function		
7	Depth Adjustable	Bari / Dimension	Laser scribe setting auto change	Programmable	Basically programme control and micrometer need to adjust for focusing
8	Pitch Adjustable	Bari / Dimension			
9	Wrong placement detection	Depth			EDI propose to use dust collector for vacum and option is whether to use vacum pump or not. Need to source out this together with EDI
10	Auto shutdown for parameter failure				Built in system
11	Drawing mc memory	Correct spec	Drawing revision control	Revision history	Drawing can't input but the cutting dimension can be input
12	Scribing Programme access code		Programme history LOG	Programme change history	Will have discussion with EDI software supplier
13	Table design	Productivity	Max utlization		Continue discussion on 21/03
14	Laser Beam adjustment	Depth	Product change vs beam strength		Need to adjust by micrometer. Micrometer need to change to metric type
15	Daily PM update				Will have discussion with EDI software supplier
16	Machine condition check	Critical parameter	System self check and confirm	SPC Graph	Need to reconfirm with EDI 21/03
17	OS upgrade to Window 7				OS is Window7
18	Downtime Record	Machine stopages	System auto capture the machine stopages	LOG report	Will have discussion with EDI software supplier
19	Safety Features		Operation & Operator safety point		Available
20	Running hours	Part change as per wear out rate	Digital timer	Digital display	Will have additional running clocks to capture the
21	Training Module - Machine operation etc	Programme & Machine control	Operation manual & system help	Soft & hardcopy	Will provide and need to send tech for training
22	Processing report	Traceability	Auto report generation after lot completion		Continue discussion on 21/03



EDI Laser Machine - Operation Process Step Control

Process	Activity	Method	Data	Responsible	Control	EDI
Initial Start (7am)	PM confirmation	Machine data capture	Numerical data	Technician	Tech ID - Barcode	This requirement need to be discuss with EDI Software supplier. Maruwa proposal is to incorporate MW PC system and other data entry. EDI engineers capability is machine making and they can't explain in detail.
Laser	Lot Registration	Input screen	PC System	Technician	Tech ID - Barcode	
	Machine Parameter check	Programme screen		Technician	Tech ID - Barcode	
	Initial buyoff	Data Input	SPC Graph	QC Operator	QC ID - Barcode	
	Machine Operation	1. Output monitoring 2. Parameter reading		Auto running		
	Downtime monitoring	1. Downtime screen 2. Product verification	Graph	Technician	Tech ID & QC ID *System allow to test run 5x	
	Lot Closure	1. Quantity 2. OPT Name 3. Yield data (Angle, etc)	Link to PC system	Leader	Leader ID - Barcode	

	Maruwa Machine frame work completed		Power reading displayed in machine monitor
	Granite plate for Linear table base		<u>Critical point #1</u> Greasing for Linear table which required for 6 months/once
	Laser Beam power unit		<u>Critical point #2</u> Laser lense - need to have spare due to tear and wear
	Table flatness confirmation (Tools will be supplied by EDI)		<u>Critical point #1</u> Filter's need to be cleaned base from machine environment
	Table unit adjustment method		Purge - need to input N2 gas to ensure low humidity
	Vacum table (proposal)		Alignment confirmation method
	4 Laser head for MMSB machine		Propose manual pick and place for speedy input
	Dust Collector outlet. *Propose to use this for table vacuum also		